

RAMSAR, Iran, Islamic Republic of

WMO: 407320

Lat: 36.9046N

Lon: 50.6840E

Elev: -20

StdP: 101.57

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	1.3	2.8	-1.6	3.3	3.6	-0.1	3.7	4.4	12.4	7.7	9.7	9.1	1.0	240	0.354

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.7	31.6	25.6	30.8	25.4	30.0	25.0	26.8	30.3	26.2	29.6	25.6	29.1	4.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	21.2	29.8	25.1	20.1	29.1	24.2	19.1	28.4	84.6	30.3	81.5	29.7	78.6	29.2	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	7.2	5.8	-0.4	33.6	1.3	2.1	-1.3	35.1	-2.1	36.4	-2.8	37.6	-3.7	39.1
			-1.3	27.7	1.3	0.9	-2.2	28.3	-3.0	28.8	-3.7	29.3	-4.6	29.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.8	8.2	7.9	10.1	13.4	19.0	23.7	26.0	26.6	23.5	19.1	13.8	10.2
	DBStd	7.16	2.45	2.51	2.70	2.77	2.64	2.24	1.84	1.78	2.25	2.59	2.85	2.50
	HDD10.0	196	64	65	30	4	0	0	0	0	0	0	4	28
	HDD18.3	1449	313	291	258	151	24	1	0	0	1	22	137	253
	CDD10.0	2693	9	7	32	105	278	410	497	513	406	283	119	33
	CDD18.3	906	0	0	1	2	44	161	239	255	156	46	2	0
	CDH23.3	7018	1	0	9	10	109	1053	2238	2556	933	103	5	1
	CDH26.7	1838	0	0	3	3	7	203	643	818	151	9	1	0
(14) Wind	WSAvg	1.9	1.8	1.9	2.2	2.0	2.0	2.1	2.2	2.0	1.9	1.8	1.7	1.7
(15) Precipitation	PrecAvg	1206	82	93	93	69	49	56	62	47	153	235	161	106
	PrecMax	1697	181	181	148	162	126	245	141	139	357	696	348	239
	PrecMin	731	42	11	1	2	17	1	8	0	11	0	21	18
	PrecStd	281	33	47	35	45	28	52	43	38	97	179	83	60
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	19.1	23.1	24.1	27.4	31.1	32.6	32.9	30.3	27.8	23.9	19.9
		MCWB	11.8	11.4	13.5	16.8	21.5	24.9	26.4	26.2	25.4	22.2	17.7	14.2
	2%	DB	14.4	14.9	18.0	20.8	25.8	29.9	31.5	31.8	29.2	25.6	20.9	17.0
		MCWB	11.1	10.3	12.7	16.5	21.3	24.1	25.4	25.9	24.7	21.7	17.6	13.3
	5%	DB	13.0	12.3	15.2	19.0	24.3	28.9	30.8	31.0	28.2	24.2	19.2	15.1
		MCWB	10.4	9.8	12.3	15.8	20.8	23.5	25.2	25.8	24.1	20.9	16.3	12.5
	10%	DB	11.9	11.2	13.2	17.4	23.2	27.8	29.8	30.1	27.2	23.2	18.1	14.1
		MCWB	9.8	9.3	11.2	15.0	20.0	23.2	24.7	25.4	23.4	20.1	15.6	11.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	12.2	15.2	18.4	23.2	26.2	27.9	27.8	26.5	23.6	19.2	15.1
		MCDB	15.6	16.3	21.2	21.4	26.0	29.7	31.5	31.3	29.6	26.3	21.7	18.0
	2%	WB	11.6	11.0	13.6	17.1	22.1	25.1	26.7	27.0	25.5	22.3	17.8	13.8
		MCDB	14.0	13.7	16.7	20.0	24.7	28.9	30.1	30.6	28.6	24.8	20.2	16.1
	5%	WB	10.7	10.2	12.4	16.0	21.1	24.3	25.9	26.5	24.7	21.4	16.9	12.9
		MCDB	12.6	11.9	14.9	18.3	23.6	27.8	29.3	30.0	27.7	23.7	18.7	14.9
	10%	WB	10.0	9.7	11.5	15.2	20.4	23.6	25.2	25.9	23.9	20.6	15.9	12.1
		MCDB	11.5	11.0	13.4	17.2	22.8	26.9	28.7	29.4	26.8	22.9	17.8	13.9
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	5.4	5.0	5.0	5.4	5.7	5.7	5.7	5.3	5.8	6.0	6.1
		MCDBR	8.1	8.2	9.7	8.3	7.0	6.9	6.9	6.8	6.3	7.0	7.6	8.0
		MCWBR	5.9	5.4	5.5	4.8	4.6	4.1	4.2	4.6	4.1	4.8	5.4	5.8
	5% WB	MCDBR	7.1	6.8	8.4	7.1	6.5	6.3	6.1	6.3	6.0	6.6	6.6	7.3
		MCWBR	5.4	4.7	5.1	4.6	4.7	4.3	4.3	4.6	4.4	4.9	5.1	5.6
(40) Clear-Sky Solar Irradiance	taub		0.399	0.450	0.511	0.596	0.588	0.592	0.644	0.611	0.572	0.526	0.446	0.402
	taud		2.140	2.016	1.905	1.777	1.837	1.860	1.779	1.846	1.903	1.970	2.091	2.147
	Ebn at Noon		755	760	750	712	727	723	681	693	691	677	695	720
	Edh at Noon		116	148	181	217	208	203	219	201	180	154	122	109
(44) All-Sky Solar Radiation	RadAvg		2.13	2.71	3.65	4.75	6.13	6.97	6.64	6.14	4.93	3.48	2.32	1.86
	RadStd		0.21	0.33	0.39	0.60	0.37	0.37	0.39	0.49	0.27	0.32	0.30	0.25

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.33	N/A	N/A	+0.86	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	+0.30	N/A	N/A	+0.70	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page